

1.3: Rates of Change and Behavior of Graphs

Definition. (Rate of Change)

A rate of change describes how an output quantity changes relative to the change in the input quantity. The units on a rate of change are “output units per input units”. The **average rate of change** between two input values is the total change of the function values divided by the input values.

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Example. Suppose the following table contains temperatures recorded through the day.

h	7 AM	9 AM	11 AM	1 PM	3 PM
$T(h)$	$78^\circ F$	$81^\circ F$	$87^\circ F$	$93^\circ F$	$98^\circ F$

Find the average rate of change between

7 AM and 9 AM

9 AM and 11 AM

7 AM and 3 PM

$$\frac{81 - 78}{9 - 7} = \boxed{\frac{3}{2}^\circ F/hr}$$

$$\frac{87 - 81}{11 - 9} = \frac{6}{2}^\circ F/hr$$

$$= \boxed{3^\circ F/hr}$$

$$\frac{98 - 78}{15 - 7} = \frac{20}{8}^\circ F/hr$$

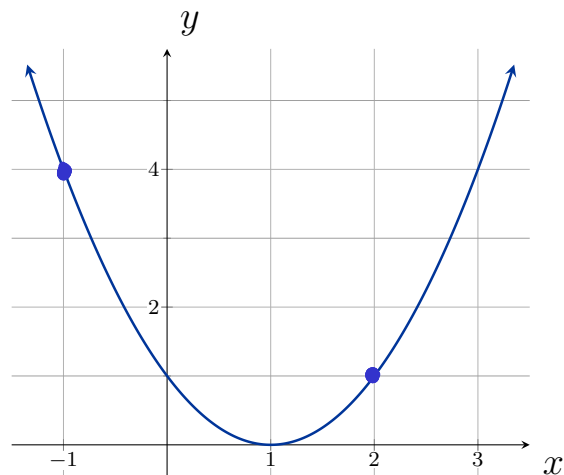
$$= \boxed{\frac{5}{2}^\circ F/hr}$$

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Example. Using the following graph, determine the average rate of change on the interval $[-1, 2]$

$$\frac{f(2) - f(-1)}{2 - (-1)} = \frac{1 - 4}{2 + 1} = \frac{-3}{3} = -1$$



Example. Find the average rate of change of $f(x) = x^2 - \frac{1}{x}$ on the interval $[2, 4]$.

$$\frac{[f(4)] - [f(2)]}{4 - 2} = \frac{\left[4^2 - \frac{1}{4}\right] - \left[2^2 - \frac{1}{2}\right]}{2}$$

$$= \frac{16 - \frac{1}{4} - 4 + \frac{1}{2}}{2}$$

$$= \frac{12.25}{2}$$

$$= 6.125 = \frac{49}{8}$$

Example. Find the expression for the average rate of change of $g(t) = t^2 + 3t + 1$ on the interval $[0, a]$.

$$\begin{aligned}\frac{g(t) - g(a)}{t - a} &= \frac{[t^2 + 3t + 1] - [a^2 + 3a + 1]}{t - a} = \frac{t^2 + 3t + 1 - a^2 - 3a - 1}{t - a} \\ &= \frac{(t^2 - a^2) + 3t - 3a}{t - a} = \frac{(t - a)(t + a) + 3(t - a)}{t - a} = \boxed{t + a + 3}\end{aligned}$$

Example. Find the expression for the average rate of change of $j(t) = \frac{1}{t+3}$ on the interval $[1, 1+h]$.

$$\frac{j(1+h) - j(1)}{(1+h) - 1} = \frac{\frac{1}{1+h+3} - \frac{1}{1+3}}{h} = \frac{\left(\frac{4}{4}\right)\frac{1}{4+h} - \frac{1}{4}\left(\frac{4+h}{4+h}\right)}{\frac{h}{1}} = \frac{\cancel{4} - \cancel{(4+h)}}{h \cdot 4(4+h)}$$

$$= \frac{-\cancel{h}}{4\cancel{h}(4+h)}$$

$$= \boxed{\frac{-1}{4(4+h)}, \quad h \neq 0}$$

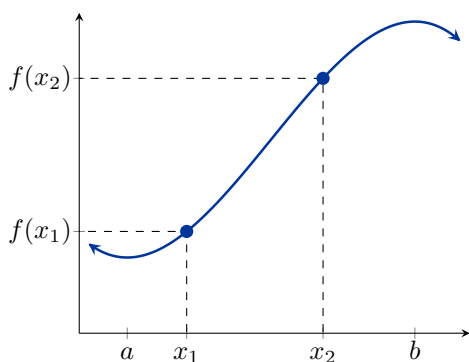
Example. Find the expression for the average rate of change of $g(x) = \sqrt{x-1}$ on the interval $[x, x+h]$.

$$\begin{aligned} \frac{f(x+h) - f(x)}{(x+h) - x} &= \frac{\sqrt{x+h-1} - \sqrt{x-1}}{h} \left(\frac{\sqrt{x+h-1} + \sqrt{x-1}}{\sqrt{x+h-1} + \sqrt{x-1}} \right) = \frac{(\sqrt{x+h-1})^2 - (\sqrt{x-1})^2}{h(\sqrt{x+h-1} + \sqrt{x-1})} \\ &= \frac{\cancel{(x+h-1)} - \cancel{(x-1)}}{h(\sqrt{x+h-1} + \sqrt{x-1})} = \frac{\cancel{h}}{\cancel{h}(\sqrt{x+h-1} + \sqrt{x-1})} \\ &= \boxed{\frac{1}{\sqrt{x+h-1} + \sqrt{x-1}}, \quad h \neq 0} \end{aligned}$$

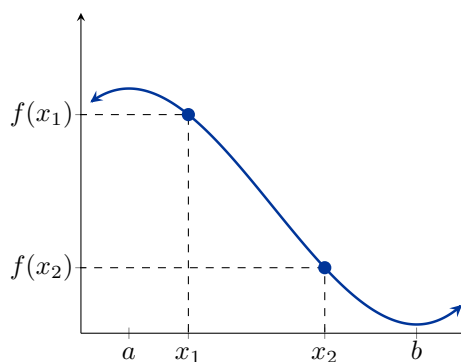
Definition.

Consider the function $f(x)$ on the interval (a, b) . Given *any* two numbers x_1 and x_2 in (a, b) where $x_1 < x_2$, we say f is

increasing if $f(x_1) < f(x_2)$



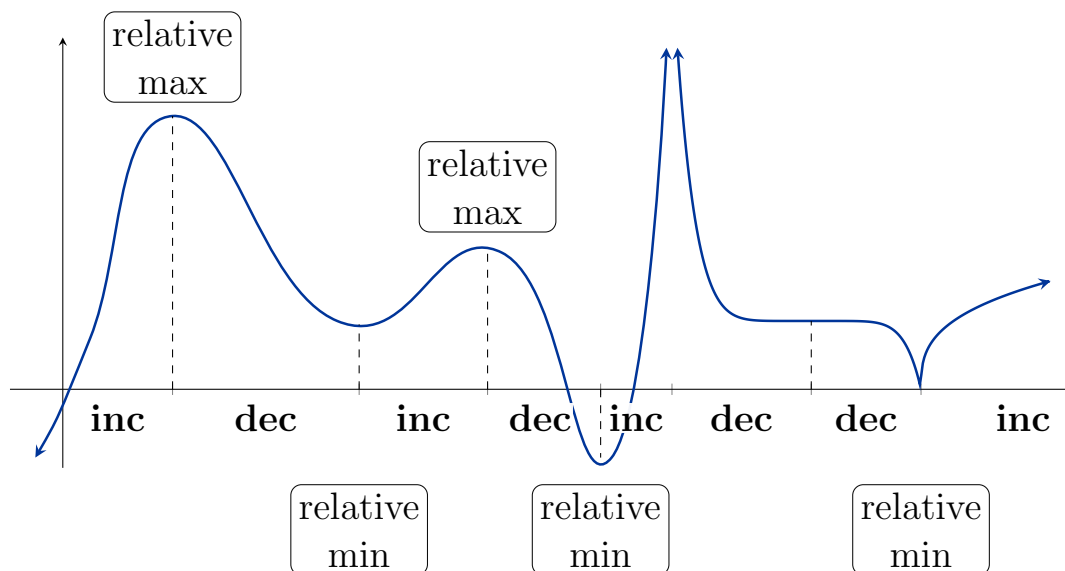
decreasing if $f(x_1) > f(x_2)$



Definition. (Relative/Local Extrema)

A function f has a

- **relative maximum** at $x = c$ if $f(c) \geq f(x)$ for every x in (a, b)
- **relative minimum** at $x = c$ if $f(c) \leq f(x)$ for every x in (a, b)

**Definition. (Absolute Extrema)**

A function f defined on a domain D has an

- **absolute maximum** at $x = c$ if $f(c) \geq f(x)$ for every x in D
- **absolute minimum** at $x = c$ if $f(c) \leq f(x)$ for every x in D

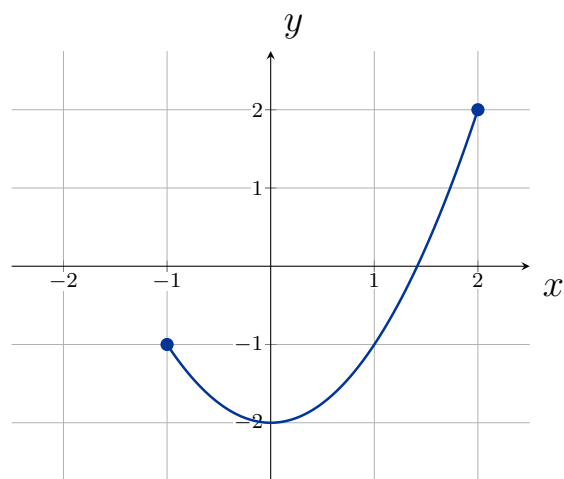
Example. Using the following graph, find any relative and absolute extrema.

rel. min of -2 at $x=0$

No rel. max

Abs. min of -2 at $x=0$

Abs. max of 2 at $x=2$



Example. Using the following graph, find any relative and absolute extrema.

rel. min of 0 at $x=0$

rel. max of 16 at $x=\pm 2$

Abs. min of -9 at $x=3$

Abs. max of 16 at $x=\pm 2$

